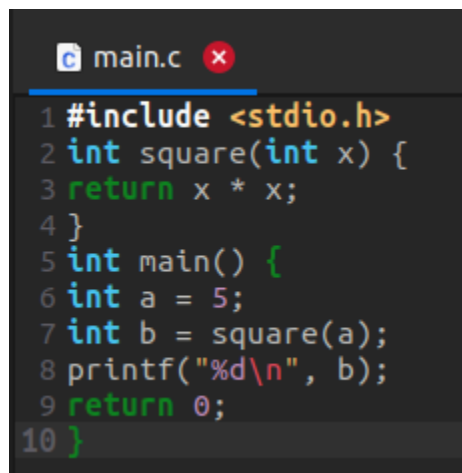


1 часть

```
aboba@tflc:~/Desktop$ sudo apt install clang llvm
[sudo] password for aboba:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binfmt-support clang-14 cpp-11 g++-11 gcc-11 gcc-11-base gcc-12-base
  icu-devtools lib32gcc-s1 lib32stdc++6 libasan6 libatomic1 libc-dev-bin libc6
  libc6-dbg libc6-dev libc6-i386 libcc1-0 libclang-common-14-dev
  libclang-cpp14 libclang1-14 libffi-dev libgcc-11-dev libgcc-s1 libgfortran5
  libgomp1 libicu-dev libitm1 libllvm14 liblsan0 libncurses-dev libobjc-11-dev
  libobjc4 libpfm4 libquadmath0 libstdc++-11-dev libstdc++6 libtinfo-dev
  libtsan0 libubsan1 libxml2 libxml2-dev libz3-4 libz3-dev llvm-14 llvm-14-dev
  llvm-14-linker-tools llvm-14-runtime llvm-14-tools llvm-runtime
  python3-pygments
Suggested packages:
  clang-14-doc gcc-11-locales g++-11-multilib gcc-11-doc gcc-11-multilib
  glibc-doc icu-doc ncurses-doc libstdc++-11-doc pkg-config llvm-14-doc
  python-pygments-doc ttf-bitstream-vera
Recommended packages:
  libnss-nis libnss-nisplus
The following NEW packages will be installed:
  binfmt-support clang clang-14 icu-devtools lib32gcc-s1 lib32stdc++6
  libc6-i386 libclang-common-14-dev libclang-cpp14 libclang1-14 libffi-dev
  libicu-dev libllvm14 libncurses-dev libobjc-11-dev libobjc4 libpfm4
  libtinfo-dev libxml2-dev libz3-4 libz3-dev llvm llvm-14 llvm-14-dev
  llvm-14-linker-tools llvm-14-runtime llvm-14-tools llvm-runtime
  python3-pygments
The following packages will be upgraded:
  cpp-11 g++-11 gcc-11 gcc-11-base libasan6 libatomic1
  libc-dev-bin libc6 libc6-dbg libc6-dev libcc1-0 libgcc-11-dev libgcc-s1
  libgfortran5 libgomp1 libitm1 liblsan0 libquadmath0 libstdc++-11-dev
  libstdc++6 libtsan0 libubsan1 libxml2
24 upgraded, 29 newly installed, 0 to remove and 378 not upgraded.
Need to get 177 MB/201 MB of archives.
After this operation, 705 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://ru.archive.ubuntu.com/ubuntu jammy/main amd64 binfmt-support amd64 2.2.1-2 [55,8 kB]
```

Рисунок 1 – Установка и подготовка среды



```
1 #include <stdio.h>
2 int square(int x) {
3     return x * x;
4 }
5 int main() {
6     int a = 5;
7     int b = square(a);
8     printf("%d\n", b);
9     return 0;
10 }
```

Рисунок 2 – Исходный код

```

| -ParmVarDecl 0x2a7b6d88 <col:20, col:32> col:32 __old 'const char *'
| -ParmVarDecl 0x2a7b6e08 <col:39, col:51> col:51 __new 'const char *'
| -NoThrowAttr 0x2a7b6f90 </usr/include/x86_64-linux-gnu/sys/cdefs.h:79:35>
| -FunctionDecl 0x2a7b7280 </usr/include/stdio.h:158:1, /usr/include/x86_64-linux-gnu/sys/cdefs.h:79:54> /usr/include/stdio.h:158:12 renameat 'int (int, const
char *, int, const char *)' extern
| -ParmVarDecl 0x2a7b7000 <col:22, col:26> col:26 __oldfd 'int'
| -ParmVarDecl 0x2a7b7080 <col:35, col:47> col:47 __old 'const char *'
| -ParmVarDecl 0x2a7b7100 <col:54, col:58> col:58 __newfd 'int'
| -ParmVarDecl 0x2a7b7180 <line:159:8, col:20> col:20 __new 'const char *'
| -NoThrowAttr 0x2a7b7340 </usr/include/x86_64-linux-gnu/sys/cdefs.h:79:35>
| -FunctionDecl 0x2a7b74a0 </usr/include/stdio.h:178:1, col:34> col:12 fclose 'int (FILE *)' extern
| -ParmVarDecl 0x2a7b73a8 <col:20, col:26> col:26 __stream 'FILE *'
| -FunctionDecl 0x2a7b7638 <line:188:1, /usr/include/x86_64-linux-gnu/sys/cdefs.h:281:58> /usr/include/stdio.h:188:14 tmpfile 'FILE *(void)' extern
| -RestrictAttr 0x2a7b76d8 </usr/include/x86_64-linux-gnu/sys/cdefs.h:281:47> malloc
| -FunctionDecl 0x2a7b7be90 </usr/include/stdio.h:205:1, /usr/include/x86_64-linux-gnu/sys/cdefs.h:79:54> /usr/include/stdio.h:205:14 tmpnam 'char *(char *)' ex
tern
| -ParmVarDecl 0x2a7bbd90 <col:22, col:35> col:26 'char *': 'char *'
| -NoThrowAttr 0x2a7bbf38 </usr/include/x86_64-linux-gnu/sys/cdefs.h:79:35>
| -FunctionDecl 0x2a7bc088 </usr/include/stdio.h:210:1, /usr/include/x86_64-linux-gnu/sys/cdefs.h:79:54> /usr/include/stdio.h:210:14 tmpnam_r 'char *(char *)'
extern
| -ParmVarDecl 0x2a7bbff0 <col:24, col:41> col:29 __s 'char *': 'char *'
| -NoThrowAttr 0x2a7bc130 </usr/include/x86_64-linux-gnu/sys/cdefs.h:79:35>
| -FunctionDecl 0x2a7bc300 </usr/include/stdio.h:222:1, /usr/include/x86_64-linux-gnu/sys/cdefs.h:281:58> /usr/include/stdio.h:222:14 tmpnam 'char *(const cha
r *, const char *)' extern
| -ParmVarDecl 0x2a7bc1a0 <col:23, col:35> col:35 __dir 'const char *'
| -ParmVarDecl 0x2a7bc220 <col:42, col:54> col:54 __pfx 'const char *'
| -NoThrowAttr 0x2a7bc3b0 </usr/include/x86_64-linux-gnu/sys/cdefs.h:79:35>
| -RestrictAttr 0x2a7bc408 <line:281:47> malloc
| -FunctionDecl 0x2a7bc4d8 </usr/include/stdio.h:230:1, col:34> col:12 fflush 'int (FILE *)' extern
| -ParmVarDecl 0x2a7bc440 <col:20, col:26> col:26 __stream 'FILE *'
| -FunctionDecl 0x2a7bc628 <line:239:1, col:43> col:12 fflush_unlocked 'int (FILE *)' extern
| -ParmVarDecl 0x2a7bc590 <col:29, col:35> col:35 __stream 'FILE *'
| -FunctionDecl 0x2a7bc8b8 <line:258:14> col:14 implicit fopen 'FILE *(const char *, const char *)' extern
| -ParmVarDecl 0x2a7bc9b0 <<invalid sloc>> <invalid sloc> 'const char *'
| -ParmVarDecl 0x2a7bca18 <<invalid sloc>> <invalid sloc> 'const char *'
| -BulletinAttr 0x2a7bc958 <<invalid sloc>> Implicit 838
| -FunctionDecl 0x2a7bca98 prev 0x2a7bc8b8 <col:1, /usr/include/x86_64-linux-gnu/sys/cdefs.h:281:58> /usr/include/stdio.h:258:14 fopen 'FILE *(const char *, co
nst char *)' extern

```

Рисунок 3 – Получение AST

```

main.ll x
1; ModuleID = 'main.c'
2 source_filename = "main.c"
3 target datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-f80:128-n8:16:32:64-S128"
4 target triple = "x86_64-pc-linux-gnu"
5
6 @.str = private unnamed_addr constant [4 x i8] c"%d\0A\00", align 1
7
8 ; Function Attrs: noline nounwind optnone uwtable
9 define dso_local @square(i32 noundef %0) #0 {
10     %2 = alloca i32, align 4
11     store i32 %0, i32* %2, align 4
12     %3 = load i32, i32* %2, align 4
13     %4 = load i32, i32* %2, align 4
14     %5 = mul nsw i32 %3, %4
15     ret i32 %5
16 }
17
18 ; Function Attrs: noline nounwind optnone uwtable
19 define dso_local @main() #0 {
20     %1 = alloca i32, align 4
21     %2 = alloca i32, align 4
22     %3 = alloca i32, align 4
23     store i32 0, i32* %1, align 4
24     store i32 5, i32* %2, align 4
25     %4 = load i32, i32* %2, align 4
26     %5 = call @square(i32 noundef %4)
27     store i32 %5, i32* %3, align 4
28     %6 = load i32, i32* %3, align 4
29     %7 = call @printf(i8* noundef getelementptr @inbounds ([4 x i8], [4 x i8]* @.str, i64 0, i64 0), i32 noundef %6)
30     ret i32 0
31 }
32
33 declare i32 @printf(i8* noundef, ...) #1

```

Рисунок 4 – Генерация LLVM IR

```

main_O0.ll
1; ModuleID = 'main.c'
2 source_filename = "main.c"
3 target_datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-f80:128-n8:16:32:64-S128"
4 target_triple = "x86_64-pc-linux-gnu"
5
6 @.str = private unnamed_addr constant [4 x i8] c"%d\0A\00", align 1
7
8 ; Function Attrs: noinline nounwind optnone uwtable
9 define dso_local i32 @square(i32 noundef %0) #0 {
10     %2 = alloca i32, align 4
11     store i32 %0, i32* %2, align 4
12     %3 = load i32, i32* %2, align 4
13     %4 = load i32, i32* %2, align 4
14     %5 = mul nsw i32 %3, %4
15     ret i32 %5
16 }
17
18 ; Function Attrs: noinline nounwind optnone uwtable
19 define dso_local i32 @main() #0 {
20     %1 = alloca i32, align 4
21     %2 = alloca i32, align 4
22     %3 = alloca i32, align 4
23     store i32 0, i32* %1, align 4
24     store i32 5, i32* %2, align 4
25     %4 = load i32, i32* %2, align 4
26     %5 = call i32 @square(i32 noundef %4)
27     store i32 %5, i32* %3, align 4
28     %6 = load i32, i32* %3, align 4
29     %7 = call i32 (i8*, ...) @printf(i8* noundef getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0, i64 0), i32 noundef %6)
30     ret i32 0
31 }
32
33 declare i32 @printf(i8* noundef, ...) #1
34
35 attributes #0 = { noinline nounwind optnone uwtable "frame-pointer"="all" "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cx8,+fxsr,+mmx,+sse,+sse2,+x87" "tune-cpu"="generic" }
36 attributes #1 = { "frame-pointer"="all" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cx8,+fxsr,

```

Рисунок 5 – Оптимизация IR (O0)

```

main_O0.ll  main_O2.ll
2 source_filename = "main.c"
3 target_datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-f80:128-n8:16:32:64-S128"
4 target_triple = "x86_64-pc-linux-gnu"
5
6 @.str = private unnamed_addr constant [4 x i8] c"%d\0A\00", align 1
7
8 ; Function Attrs: mustprogress norecurse nosync nounwind readnone uwtable willreturn
9 define dso_local i32 @square(i32 noundef %0) local_unnamed_addr #0 {
10     %2 = mul nsw i32 %0, %0
11     ret i32 %2
12 }
13
14 ; Function Attrs: norefree nounwind uwtable
15 define dso_local i32 @main() local_unnamed_addr #1 {
16     %1 = tail call i32 (i8*, ...) @printf(i8* noundef nonnull dereferenceable(1) getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0, i64 0), i32
noundef 25)
17     ret i32 0
18 }
19
20 ; Function Attrs: norefree nounwind
21 declare noundef i32 @printf(i8* nocapture noundef readonly, ...) local_unnamed_addr #2
22
23 attributes #0 = { mustprogress norecurse nosync nounwind readnone uwtable willreturn "frame-pointer"="none" "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cx8,+fxsr,+mmx,+sse,+sse2,+x87" "tune-cpu"="generic" }
24 attributes #1 = { norefree nounwind uwtable "frame-pointer"="none" "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cx8,+fxsr,+mmx,+sse,+sse2,+x87" "tune-cpu"="generic" }
25 attributes #2 = { norefree nounwind "frame-pointer"="none" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cx8,+fxsr,+mmx,+sse,+sse2,+x87" "tune-cpu"="generic" }
26
27 !llvm.module.flags = !{!0, !1, !2, !3}
28 !llvm.ident = !{!4}
29
30 !0 = !{i32 1, !"wchar_size", i32 4}
31 !1 = !{i32 7, !"PIC Level", i32 2}
32 !2 = !{i32 7, !"PIE Level", i32 2}
33 !3 = !{i32 7, !"uwtable", i32 1}
34 !4 = !{"!Ubuntu clang version 14.0.0-1ubuntu1.1"}

```

Рисунок 6 – Оптимизация IR (O2)

```

aboba@tflc:~/Desktop/7$ diff main_00.ll main_02.ll
8,15c8,11
< ; Function Attrs: noinline nounwind optnone uwtable
< define dso_local i32 @square(i32 noundef %0) #0 {
<   %2 = alloca i32, align 4
<   store i32 %0, i32* %2, align 4
<   %3 = load i32, i32* %2, align 4
<   %4 = load i32, i32* %2, align 4
<   %5 = mul nsw i32 %3, %4
<   ret i32 %5
---
> ; Function Attrs: mustprogress norecurse nosync nounwind readnone uwtable willreturn
> define dso_local i32 @square(i32 noundef %0) local_unnamed_addr #0 {
>   %2 = mul nsw i32 %0, %0
>   ret i32 %2
18,29c14,16
< ; Function Attrs: noinline nounwind optnone uwtable
< define dso_local i32 @main() #0 {
<   %1 = alloca i32, align 4
<   %2 = alloca i32, align 4
<   %3 = alloca i32, align 4
<   store i32 0, i32* %1, align 4
<   store i32 5, i32* %2, align 4
<   %4 = load i32, i32* %2, align 4
<   %5 = call i32 @square(i32 noundef %4)
<   store i32 %5, i32* %3, align 4
<   %6 = load i32, i32* %3, align 4
<   %7 = call i32 (i8*, ...) @printf(i8* noundef getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0, i64 0), i32 noundef %6)
---
> ; Function Attrs: norecurse nounwind uwtable
> define dso_local i32 @main() local_unnamed_addr #1 {
>   %1 = tail call i32 (i8*, ...) @printf(i8* noundef nonnull dereferenceable(1) getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0, i64 0), i32 noundef
25)
33c20,21
< declare i32 @printf(i8* noundef, ...) #1
---
> ; Function Attrs: norecurse nounwind
> declare noundef i32 @printf(i8* nocapture noundef readonly, ...) local_unnamed_addr #2
35,36c23,25
< attributes #0 = { noinline nounwind optnone uwtable "frame-pointer"="all" "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cx8,+fxsr,+mmx,+sse,+sse2,+x87" "tune-cpu"="generic" }

```

Рисунок 7 – Сравнение двух файлов

```

aboba@tflc:~/Desktop/7$ clang -O2 -S -emit-llvm main.c -o main.ll
aboba@tflc:~/Desktop/7$ opt -dot-cfg -disable-output main.ll
Writing '.square.dot'...
Writing '.main.dot'...
aboba@tflc:~/Desktop/7$ find . -name "*.dot"
./square.dot
./main.dot

```

Рисунок 8 – Генерация LLVM IR и dot-файлы

```

aboba@tflc:~/Desktop/7$ sudo apt install graphviz
[sudo] password for aboba:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libann0 libcdt5 libcgraph6 libgts-0.7-5 libgts-bin libgvc6 libgvpr2 liblab-gamut1 libpathplan4
Suggested packages:
  graphviz-doc
The following NEW packages will be installed:
  graphviz libann0 libcdt5 libcgraph6 libgts-0.7-5 libgts-bin libgvc6 libgvpr2 liblab-gamut1 libpathplan4
0 upgraded, 10 newly installed, 0 to remove and 378 not upgraded.
Need to get 3 858 kB of archives.
After this operation, 8 563 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://ru.archive.ubuntu.com/ubuntu jammy/universe amd64 libann0 amd64 1.1.2+doc-7build1 [26,0 kB]
Get:2 http://ru.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libcdt5 amd64 2.42.2-6ubuntu0.1 [21,1 kB]
Get:3 http://ru.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libcgraph6 amd64 2.42.2-6ubuntu0.1 [45,4 kB]
Get:4 http://ru.archive.ubuntu.com/ubuntu jammy/universe amd64 libgts-0.7-5 amd64 0.7.6+darcs121130-5 [164 kB]
Get:5 http://ru.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libpathplan4 amd64 2.42.2-6ubuntu0.1 [23,4 kB]
Get:6 http://ru.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libgvc6 amd64 2.42.2-6ubuntu0.1 [724 kB]
Get:7 http://ru.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libgvpr2 amd64 2.42.2-6ubuntu0.1 [192 kB]
Get:8 http://ru.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 liblab-gamut1 amd64 2.42.2-6ubuntu0.1 [1 965 kB]
Get:9 http://ru.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 graphviz amd64 2.42.2-6ubuntu0.1 [653 kB]
Get:10 http://ru.archive.ubuntu.com/ubuntu jammy/universe amd64 libgts-bin amd64 0.7.6+darcs121130-5 [44,3 kB]
Fetched 3 858 kB in 2min 32s (25,3 kB/s)
Selecting previously unselected package libann0.
(Reading database ... 299556 files and directories currently installed.)
Preparing to unpack .../0-libann0_1.1.2+doc-7build1_amd64.deb ...
Unpacking libann0 (1.1.2+doc-7build1) ...
Selecting previously unselected package libcdt5:amd64.
Preparing to unpack .../1-libcdt5_2.42.2-6ubuntu0.1_amd64.deb ...
Unpacking libcdt5:amd64 (2.42.2-6ubuntu0.1) ...
Selecting previously unselected package libcgraph6:amd64.
Preparing to unpack .../2-libcgraph6_2.42.2-6ubuntu0.1_amd64.deb ...
Unpacking libcgraph6:amd64 (2.42.2-6ubuntu0.1) ...
Selecting previously unselected package libgts-0.7-5:amd64.
Preparing to unpack .../3-libgts-0.7-5_0.7.6+darcs121130-5_amd64.deb ...
Unpacking libgts-0.7-5:amd64 (0.7.6+darcs121130-5) ...
Selecting previously unselected package libpathplan4:amd64.
Preparing to unpack .../4-libpathplan4_2.42.2-6ubuntu0.1_amd64.deb ...
Unpacking libpathplan4:amd64 (2.42.2-6ubuntu0.1) ...

```

Рисунок 9 – Установка Graphviz

```

aboba@tflc:~/Desktop/7$ dot -Tpng .main.dot -o cfg_main.png
aboba@tflc:~/Desktop/7$ dot -Tpng .square.dot -o cfg_square.png

```

Рисунок 10 – Создание файлов

```

%0:
%1 = tail call i32 @i8*, ... @printf(i8* noundef nonnull dereferenceable(1)
... getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0, i64 0), i32 noundef
... 25)
ret i32 0

```

CFG for 'main' function

Рисунок 11 – CFG main

```
%1:  
%2 = mul nsw i32 %0, %0  
ret i32 %2
```

CFG for 'square' function

Рисунок 12 – CFG square